

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular /Supplementary Winter Examination-2024

Course: B. Pharmacy

Semester: III

Subject Name: Pharmaceutical Engineering

Subject Code: BP304T

Max Mark:75

Date: 26/12/2024

Duration:3 Hr.

Instructions:

1. All questions are compulsory.
2. Figure to the right indicate full marks of that question
3. Draw Diagrams/figures wherever necessary.

Q.1

Objective Type questions (Answer All the Questions)

(10x2) = 20

- A Enlist and describe the mechanisms of size reduction.
- B State Fourier's Law and provide its equation.
- C Define Reynolds number and discuss its importance in fluid mechanics.
- D Analyze the advantages and disadvantages of plastics as construction materials in the Pharmaceutical industry.
- E Give different mechanisms of heat transfer.
- F Define, classify corrosion and provide examples to illustrate the concept.
- G Draw neat and labelled diagram for Hammer Mill.
- H Differentiate between Drying & Evaporation.
- I What is mean by Filter Aids? Give its Example
- J Enlist Factors influencing on Evaporation.

Q.2

Long Answers (Solve Any 2 out of 3)

(2x10)= 20

- A Compare and contrast fluid flow, fluid statics, and fluid dynamics. Differentiate between orifice meter and venturimeter. Describe venturimeter in detail.
- B Define & Classify distillation. Explain the principle and working of steam distillation & Molecular distillation.
- C What are the main phases of a drying curve and what happens in each phase? How does a spray dryer work, and what are its key components? along with its Advantages and disadvantages.

Q.3

Short Answers (Solve Any 7 out of 9)

(7x5)= 35

- A Describe the principle, construction, and working of a ball mill.
- B Explain principle, construction, working of perforated basket centrifuge.
- C With help of neat labelled diagram Explain principle, construction, working & uses of fluidized bed dryer.
- D Differentiate between solid mixing and liquid mixing. Describe the principle, construction of ribbon blender.
- E Explain principle, construction, working of perforated basket centrifuge.
- F Classify materials of construction and discuss various types of ferrous metals used in pharmaceuticals.
- G Define Filtration & Explain in detail factor influencing on filtration
- H Explain the Bernoulli theorem, give its significance.
- I Explain the applications of size separation and describe the working and details of a sieve shaker

BEST OF LUCK